

Revision 0: Requirements and Hazard Analysis

Group 8 - C.A.V.E: Cave Assessment and Visualization Equipment

Abdul Rahim Khan
Andrew Brink
Gokberk Yilmaz
Nicholas Trimble
Tabish Faisal

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1 Introduction

This document will define the scope of the C.A.V.E. system: what inputs it collects, what processing it does, and what outputs it creates. It will define all the variables and constants used to achieve this, as well as any notation or technical terms that are needed. There will also be a fault tree analysis (FTA) breakdown of the possible root causes of failure that will inform some of the safety requirements of the system.

2 Requirements

System Diagram

Definitions and Notation

Variables – monitored, controlled. Constants w/ nominal values.

Diagrams showing decomposition (?)

Safety/security requirements – reference hazard analysis in appendix

Requirements – required behaviour description, no design. Rationale, timing reqs, hardware, software, undesired event handling (safety), which are likely to change or aren't

Very rough sketch of what is part of our system, what is not, and what the general structure of our design is (this is definitely not final, but I think it is a good start):

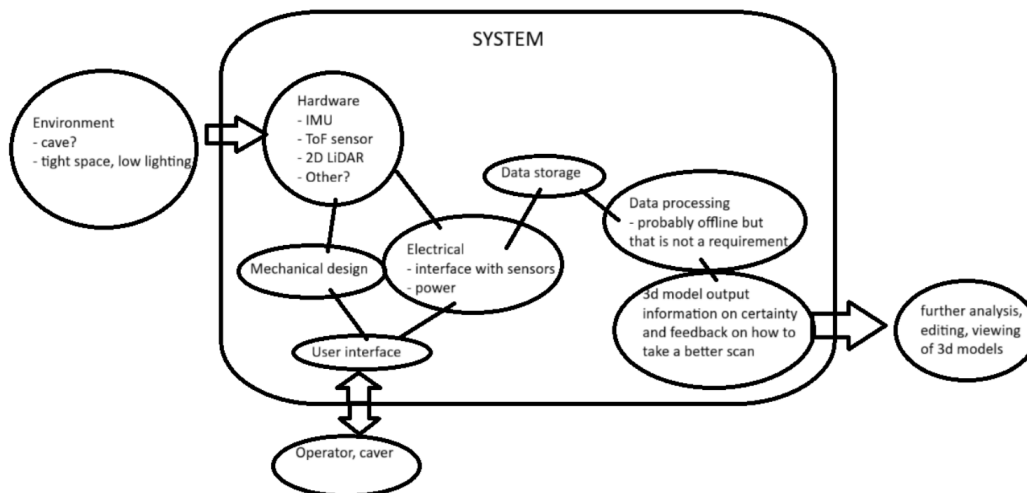


Figure 1: General structure of the system

Monitored variables:

- M_tof_data: matrices of distances to objects in environment
- M_imu_data: Orientation information from sensor (not sure if all the sensed values should be broken out here, if so that would be magnetometer, accelerometer, and gyroscope x, y, and z values) -going the split them to pad the report :) -Nicholas
- M_lidar_data: 2D Li-DAR scan

- M_start_stop: user interface button to start, stop scan
- M_battery_level: no idea if we can monitor battery levels; even if so, should this be included here?

2.1 Monitored and Controlled Variables

Monitored variables are information from the environment being input to the system, and controlled variables are the output products of the system's internal processes. The following tables list and describe these variables in the C.A.V.E. system.

Variable	Description	Unit	Purpose
M_tof_data	A single dot matrix frame captured from the environment	Array of depths and positions in metres	Capturing the contours and shape of the environment
M_imu_position	Device distance from C_START_POSN as measured by the IMU	Metres in (x,y,z) !!!I'm assuming the IMU measures an offset from its power on location	Provide reference for position change between M_tof_data captures
M_imu_attitude	Device rotation about the x, y, z axes, measured by IMU	Degrees: rotation from a reference frame	Provide reference for space frame rotation
M_imu_accel	Device acceleration in x, y, z, directions, measured by IMU	Metres/second ² tuple (x,y,z)	Additional frame translation reference
M_imu_magnet	Change in magnetic field	Check how its expressed- tesla/s?	Additional frame rotation reference
M_lidar	2D Li-DAR scan of the environment	Tuple of distance in metres and angle in degrees	Long range slice of the environment to aid convergence
M_start_stop	User toggle enabling or disabling the system	Boolean, true == on	Allow user control of starting and stopping a mapping session
M_SOC	Remaining battery percentage	% of total charge	Estimate remaining time available for mapping
C_record_status	User facing indicator to show the status of M_start_stop	Boolean	Inform user of the status of M_start_stop
C_record_remain	...	...	...

Controlled variables:

- C_record_status: user interface light indicator (stopped, recording, error, etc)

- C_record_time: measurement (minutes?) of how long recording can continue (based on battery level and amount of storage) (I don't know if this is something we can do with our hardware, it just seems like a good way to fluff up the controlled variables and the system requirements)
- C_point_cloud: point cloud created by merging all the data from the recording
- C_confidence: This could be anything from a binary "success" or "failure" indicator to a more complex indicator of which parts of the map are accurate and which ones should be scanned again in more detail. We should probably specify which one it is.

Constants:

- Transformation matrix from each sensor to the global coordinate system (based on mechanical design)
- Update rate for sensors (possibly different for each sensor)
- Storage capacity of recording computer
- Data rate for recording

Requirements:

- C_point_cloud contains a sampling of points that represent the physical surface of the environment which has been seen by the sensors (M_tof_data, M_lidar_data) while the recording is active.
- The accuracy and reliability of the data contained in C_point_cloud must be reflected by the C_confidence variable.
- C_record_status toggles between recording and standby when M_start_stop button is pressed. If there is an error which the system cannot recover from on its own, C_record_status goes to error status.
- C_record_time counts down the amount of time remaining for recording, taking the minimum of the estimated battery life (inferred using M_battery_level) and the amount of recording time based on free device storage space and data rate of recording.

Performance timing requirements:

Data sample rate from sensors is the only one I can think of

Requirements on hardware environment:

Cave proof?

Requirements on software environment:

...

3 Hazard Analysis

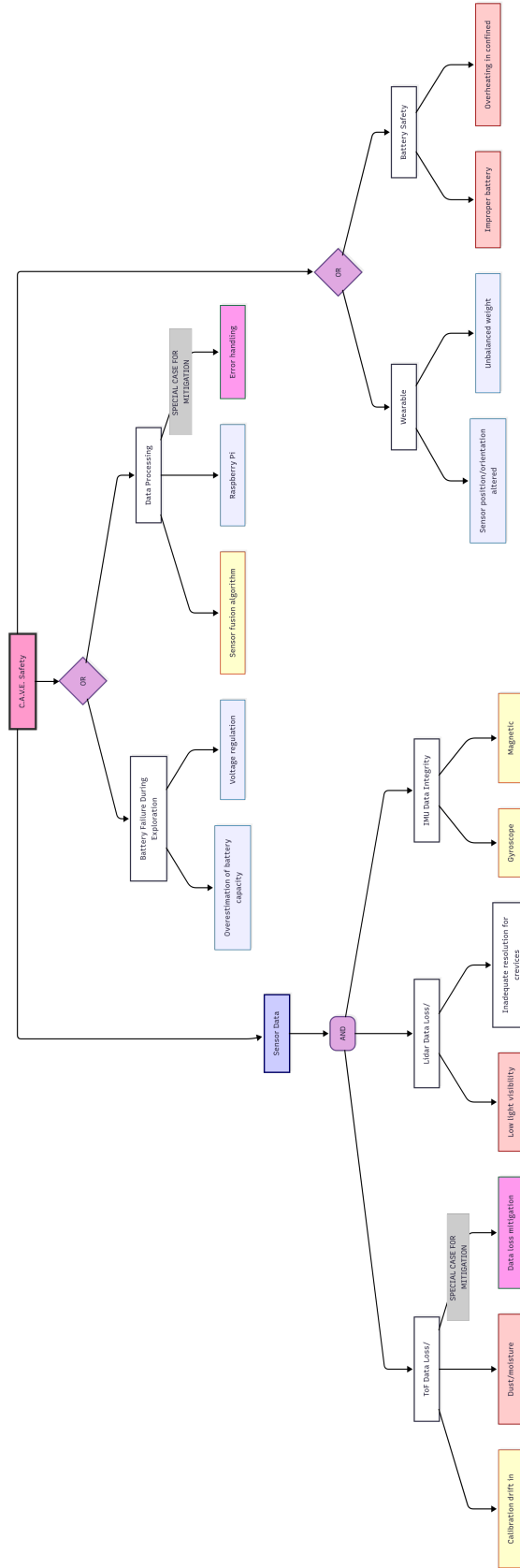


Figure 2: Fault tree analysis of the system